

LEDs: The Future of Lighting

Whether it be climate change caused by global warming or huge electricity bills of your house, LED lights are the answer to all the worries. Already being used widely, these are definitely going to replace the lamps that we use today

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Light-emitting diodes (LEDs) are small but powerful devices in terms of their diverse applications. LED lights assume greater significance in the context of need for electrical energy conservation and pollution control world over. These are used as indicators in various equipment, for lighting and decorations at homes, and in flashlights, signboards and car lights.

LED lights differ from traditional light sources in the way these produce light. In incandescent bulbs, a tungsten filament is heated by electric current until it glows or emits light. In fluorescent lamps, an electric arc excites mercury atoms, which emit ultraviolet (UV) radiation. After striking the phosphor coating on the inner side of the glass tubes, the UV radiation is converted into visible light and emitted. On the other hand, LEDs are

diodes made from semiconductor materials. This is why these are referred to as solidstate devices. They rely on indium-gallium nitride to convert electricity into photons when current flows through it.

An LED is basically a small-area source, often with extra optics added to the chip that shapes its radiation pattern. The specific wavelength or colour emitted by the LED depends on the materials used to make the diode. It depends on the composition and condition of the semiconducting material used, and can be infrared, visible or near-ultraviolet.

LED-based light technology is a subject of intense research and development. It is predicted to radically change the energy consumption patterns throughout the world over the next few years. Therefore a phenomenal amount of resources are being devoted to improve the LED technology worldwide. The US has committed a large amount of industrial policy support

for R&D on LEDs. Asia is a hot bed of cutting-edge manufacturing technology, and Europe and Canada have led the way in energy-efficient initiatives accompanied by some of the most innovative design styles.

How does an LED work?

Light generates inside the semiconductor chip when current flows across the junction of different material compositions. An LED consists of two elements of processed semiconductor material called p-type and n-type semiconductors. These two elements form a p-n junction when placed in contact with each other. The material composition of the p- and n-type semiconductors determines the colour of light generated.

Fig. 1 shows the parts of a common type of LED structure. There are major differences between LEDs and other diode types. An LED has a transparent package, allowing visible or IR energy to pass through. Also, the LED has a large p-n junction area whose shape is tailored to the application.

The device shown in Fig. 1 is an AlGaInP LED because the semiconductor layers are aluminium (Al), gallium (Ga), indium (In) and phosphate (P). It is illuminated with the movement of electrons in the semiconductor material, when an electric current passes through it.

Connecting, soldering and testing LEDs

LEDs must be connected the correct way round. A simple way to identify cathode is the short lead, and the body of round LEDs may be slightly flat towards cathode. If you see inside an LED, you will notice that the cathode is the larger electrode.

